



BOARD QUESTION PAPER : JULY 2024

MATHEMATICS AND STATISTICS

Time: 3 Hrs.

Max. Marks: 80

General instructions:The question paper is divided into **FOUR** sections.

- (1) **Section A:** Q. 1 contains **Eight** multiple choice type of questions, each carrying **Two** marks.
Q. 2 contains **Four** very short answer type questions, each carrying **One** mark.
- (2) **Section B:** Q. 3 to Q. 14 contain **Twelve** short answer type questions, each carrying **Two** marks. (Attempt any **Eight**)
- (3) **Section C:** Q. 15 to Q. 26 contain **Twelve** short answer type questions, each carrying **Three** marks. (Attempt any **Eight**)
- (4) **Section D:** Q. 27 to Q. 34 contain **Eight** long answer type questions, each carrying **Four** marks. (Attempt any **Five**)
- (5) Use of log table is allowed. Use of calculator is not allowed.
- (6) Figures to the right indicate full marks.
- (7) Use of graph paper is not necessary. Only rough sketch of graph is expected.
- (8) For each multiple choice type of question, only the first attempt will be considered for evaluation.
- (9) Start answer to each section on a new page.

SECTION – A**Q.1. Select and write the correct answer for the following multiple choice type of questions:** [16]

- i. $\cos\left[\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{2}\right)\right] = \underline{\hspace{2cm}}$.
 (A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\pi}{4}$ (2)
- ii. If θ is the angle between two vectors $\vec{a}$ and $\vec{b}$ and $|\vec{a} + \vec{b}| = |\vec{a} \times \vec{b}|$ then θ is equal to _____.
 (A) 0 (B) $\frac{\pi}{4}$ or $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π or $\frac{\pi}{6}$ (2)
- iii. The angle between the lines $\vec{r} = (\hat{i} + 2\hat{j} - 3\hat{k}) + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k})$ and $\vec{r} = (5\hat{i} - 2\hat{j} + 7\hat{k}) + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$ is _____.
 (A) $\cos^{-1}\left(\frac{17}{21}\right)$ (B) $\cos^{-1}\left(\frac{20}{21}\right)$ (C) $\cos^{-1}\left(\frac{18}{21}\right)$ (D) $\cos^{-1}\left(\frac{19}{21}\right)$ (2)
- iv. The perpendicular distance of the plane $\vec{r} \cdot (2\hat{i} + 3\hat{j} - \hat{k}) = 5$, from the origin is _____.
 (A) $\frac{5}{\sqrt{14}}$ units (B) $\frac{5}{14}$ units (C) 5 units (D) $\frac{\sqrt{14}}{5}$ units (2)
- v. If $x = e^{\frac{x}{y}}$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.
 (A) $1 - \frac{y}{x}$ (B) $1 + \frac{y}{x}$ (C) $\frac{x-y}{x \log x}$ (D) $\frac{x+y}{x \log x}$ (2)
- vi. $y = c^2 + \frac{c}{x}$ is solution of _____.
 (A) $x^4 \left(\frac{dy}{dx}\right)^2 - x \frac{dy}{dx} - y = 0$ (B) $x^2 \left(\frac{dy}{dx}\right)^2 + y = 0$
 (C) $x^3 \left(\frac{d^2y}{dx^2}\right) - x \frac{dy}{dx} + y = 0$ (D) $x \frac{d^2y}{dx^2} = 4y$ (2)





- vii. Given that $X \sim B(n, p)$. If $n = 10$ and $p = 0.4$ then $E(X)$ and $\text{Var}(X)$ respectively are _____.
 (A) 4, 0.24 (B) 0.4, 0.24 (C) 4, 2.4 (D) 3, 0.24 (2)
- viii. The approximate value of $\tan(44^\circ 30')$, given that $1^\circ = 0.0175^\circ$, is _____.
 (A) 0.8952 (B) 0.9528 (C) 0.9285 (D) 0.9825 (2)

Q.2. Answer the following questions:

- i. Find the combined equation of the pair of lines $2x + y = 0$ and $3x - y = 0$ (1)
- ii. Find the value of $\sin^{-1}\left(\sin\frac{5\pi}{3}\right)$ (1)
- iii. Evaluate: $\int \frac{5^x}{3^x} dx$ (1)
- iv. Write the integrating factor (I.F.) of the differential equation $\frac{dy}{dx} + y = e^{-x}$. (1)

SECTION – B**Attempt any EIGHT of the following questions:**

Q.3. If the statements p, q are true statements and r, s are false statements, then determine the truth value of the statement pattern:

$$(q \wedge r) \vee (\sim p \wedge s) \quad (2)$$

Q.4. Find the inverse of matrix A by elementary row transformations, where $A = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$ (2)

Q.5. Find the polar co-ordinates of the point whose Cartesian co-ordinates are $(1, -\sqrt{3})$. (2)

Q.6. Find the acute angle between the lines represented by $xy + y^2 = 0$ (2)

Q.7. Using the truth table, show that the statement pattern $p \rightarrow (q \rightarrow p)$ is a tautology. (2)

Q.8. If $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$ then find the value of x , where $0 < 3x < 1$. (2)

Q.9. Find the points on the curve given by $y = x^3 - 6x^2 + x + 3$ where the tangents are parallel to the line $y = x + 5$. (2)

Q.10. Evaluate: $\int \frac{e^x(1+x)dx}{\sin^2(xe^x)}$. (2)

Q.11. The displacement of a particle at a time t is given by $s = 2t^3 - 5t^2 + 4t - 3$. Find the time when acceleration is 14 ft/sec^2 . (2)

Q.12. Evaluate: $\int_0^{\pi/4} \sqrt{1 + \sin 2x} dx$ (2)

Q.13. The probability distribution of X is as follows:

$X = x$	0	1	2	3	4
$P(X = x)$	0.1	k	$2k$	$2k$	k

- Find (a) k
 (b) $P(X < 2)$ (2)

Q.14. Find the particular solution of:

$$r \frac{dr}{d\theta} + \cos \theta = 5 \text{ at } r = \sqrt{2} \text{ and } \theta = 0 \quad (2)$$





SECTION – C

Attempt any EIGHT of the following questions:

[24]

Q.15. In $\triangle ABC$, if $a \cos A = b \cos B$ then prove that the triangle is either a right angled or an isosceles triangle. (3)

Q.16. Are the four points $A(1, -1, 1)$, $B(-1, 1, 1)$, $C(1, 1, 1)$ and $D(2, -3, 4)$ co-planar? Justify your answer. (3)

Q.17. Find the difference between the slopes of the lines given by $(\tan^2 \theta + \cos^2 \theta)x^2 - 2xy \tan \theta + (\sin^2 \theta)y^2 = 0$ (3)

Q.18. Find the vector equation of the line passing through the point $(\hat{i} + 2\hat{j} + 3\hat{k})$ and perpendicular to the vectors $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} - \hat{j} + \hat{k}$. (3)

Q.19. Let $\vec{a}$ and $\vec{b}$ be non-collinear vectors. If vector $\vec{r}$ is co-planar with $\vec{a}$ and $\vec{b}$ then prove that there exists unique scalars t_1 and t_2 such that $\vec{r} = t_1 \vec{a} + t_2 \vec{b}$. Hence find t_1 and t_2 for $\vec{r} = \hat{i} + \hat{j}$, $\vec{a} = 2\hat{i} - \hat{j}$, $\vec{b} = \hat{i} - 2\hat{j}$. (3)

Q.20. Find the equation of the plane passing through the intersection of the planes $x + 2y + 3z + 4 = 0$ and $4x + 3y + 2z + 1 = 0$ and the origin. (3)

Q.21. Find $\frac{dy}{dx}$ if $y = \tan^{-1} \left(\sqrt{\frac{3-x}{3+x}} \right)$ (3)

Q.22. Find the approximate value of $f(x) = x^3 + 5x^2 - 2x + 3$ at $x = 1.98$. (3)

Q.23. Evaluate: $\int \frac{\sin(x+a)}{\cos(x-b)} dx$ (3)

Q.24. Solve the differential equation $dr + (2r \cot \theta + \sin 2\theta) d\theta = 0$ (3)

Q.25. Let $X \sim B(10, 0.2)$. Find

(a) $P(X = 1)$

(b) $P(X \geq 1)$

Q.26. Find the expected value, variance and standard deviation of r.v. X whose p.m.f. is given as:

$X = x$	1	2	3
$P(X)$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{2}{5}$

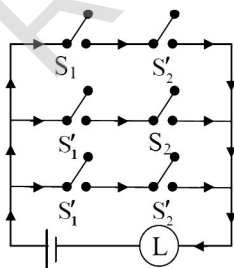
(3)

SECTION – D

Attempt any FIVE of the following questions:

[20]

Q.27. Give an alternative arrangement for the following circuit, so that the new circuit has minimum switches:



(4)

Q.28. If $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ then find A^{-1} by Adjoint method. (4)





Q.29. In $\triangle ABC$, D and E are points on BC and AC respectively such that $BD = 2DC$ and $AE = 3EC$. Let P be the point of intersection of AD and BE. Find ratio $\frac{BP}{PE}$ using the vector method. (4)

Q.30. A firm manufactures two products A and B on which profit earned per unit are ₹3 and ₹4 respectively. Each product is processed on two machines M_1 and M_2 . The product A requires one minute of processing time on M_1 and two minutes on M_2 , while product B requires one minute on M_1 and one minute on M_2 . Machine M_1 is available for use not more than 450 minutes, while M_2 is available for 600 minutes during any working day. Find the number of units of products A and B to be manufactured to get maximum profit. (4)

Q.31. If $y = f(u)$ is a differentiable function of u and $u = g(x)$ is differentiable function of x such that the composite function $y = f[g(x)]$ is a differentiable function of x then prove that: $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$.

Hence find $\frac{d}{dx} \left(\frac{1}{\sqrt{\sin x}} \right)$ (4)

Q.32. Evaluate: $\int x^2 \sin 3x dx$ (4)

Q.33. Prove that:

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx, \text{ if } f \text{ is an even function}$$

$$= 0, \text{ if } f \text{ is an odd function}$$

Hence find the value of $\int_{-1}^1 \tan^{-1} x dx$ (4)

Q.34. Find the area of the ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Hence write area of $\frac{x^2}{25} + \frac{y^2}{16} = 1$ (4)

